



Lecture-4

Hierarchical Approach to Conceptual Design

11-08-2026

Synthesis Step



Synthesis Step

- ① Eliminate differences in molecular types
- ② Distribute chemicals by matching *sources* and *sinks*
- ③ Eliminate differences in composition
- ④ Eliminate differences in temperature, pressure and phase
- ⑤ Integrate tasks (combine *tasks* into *unit operations*)

Process Operation

Chemical reaction

Mixing

Separation

Temperature, pressure and phase change

Source Ref: Seider et al., 2010

Example of Process Synthesis



- Vinyl Chloride Manufacture
 - Used for the manufacture of polyvinyl chloride
 - An important polymer that is widely used for rigid plastic piping, fittings, and similar products
- Vinyl chloride monomer is extremely toxic substance, and therefore industrial plants that manufacture it or process must be designed carefully
- In this example, only the production of the monomer is considered

Step- 1: Eliminate differences in molecular types



Chemicals participating in VC Manufacture:

Chemical	Molecular weight	Chemical formula	Chemical structure
Acetylene	26.04	C_2H_2	$H-C \equiv C-H$
Chlorine	70.91	Cl_2	$Cl-Cl$
1,2-Dichloroethane	98.96	$C_2H_4Cl_2$	<pre> Cl Cl H-C-C-H H H </pre>
Ethylene	28.05	C_2H_4	<pre> H H \ / C = C / \ H H </pre>
Hydrogen chloride	36.46	HCl	$H-Cl$
Vinyl chloride	62.50	C_2H_3Cl	<pre> H Cl \ / C = C / \ H H </pre>

Source Ref: Seider et al., 2010

Step- 1: Eliminate differences in molecular types

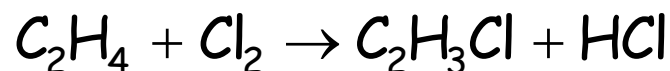


- For newer substances, the design team often begins to carry out process synthesis as the data are emerging from laboratory
- The challenge in these cases,
 - To guide the chemists away from those reaction paths that lead to processes that are costly to build and operate, and
 - To arrive at design as quickly as possible, in time to capture the market before a competitive process or chemical is developed by another company

Selection of pathway to VCM



① Direct chlorination of ethylene:



Advantages:

- Attractive solution to the specific problem denoted as **Alternative 2** in analysis of primitive problem.
- Occurs spontaneously at a few hundred °C.

Disadvantages:

- Does not give a high yield of VC without simultaneously producing large amounts of by-products such as dichloroethylene
- Half of the expensive chlorine is consumed to produce HCl by-product, which may not be sold easily.

Selection of pathway to VCM



② Hydrochlorination of acetylene:



Advantages:

- This exothermic reaction is a potential solution for the specific problem denoted as **Alternative 3**. It provides a good conversion (98%) of C_2H_2 to VC at 150°C in the presence of HgCl_2 catalyst impregnated in activated carbon at atmospheric pressure.
- These are fairly moderate reaction conditions, and hence, this reaction deserves further study.

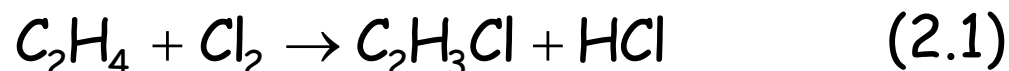
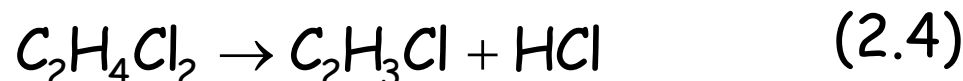
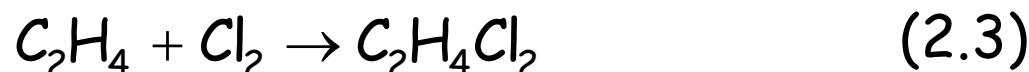
Disadvantages:

- Flammability limits of C_2H_2 (2.5 → 100%)

Selection of pathway to VCM



③ Thermal cracking of $C_2H_4Cl_2$ from chlorination of C_2H_4 :



Advantages:

- Conversion of ethylene to 1,2-dichloroethane in exothermic reaction (2.3) is $\approx 98\%$ at $90^\circ C$ and 1 atm with a Friedel-Crafts catalyst such as $FeCl_3$. This intermediate is converted to vinyl chloride by thermal cracking according to the endothermic reaction (2.4), which occurs spontaneously at $500^\circ C$ with conversions as high as 65% (Alternative 2).

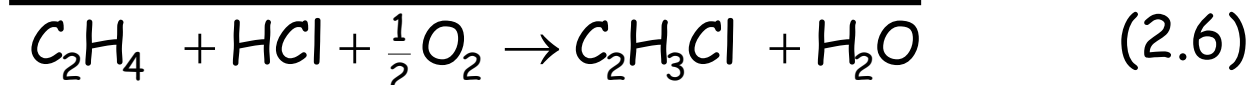
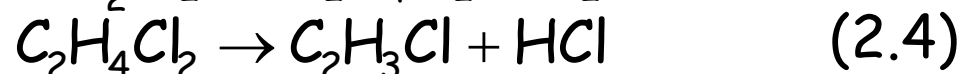
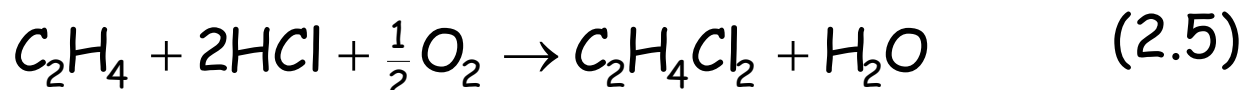
Disadvantage:

- Half of the expensive chlorine is consumed to produce HCl by-product, which may not be sold easily.

Selection of pathway to VCM



④ Thermal Cracking of $C_2H_4Cl_2$ from Oxychlorination of C_2H_4 :



Advantages:

- Highly exothermic reaction (2.5) achieves a 95% conversion to $C_2H_4Cl_2$ in the presence of $CuCl_2$ catalyst, followed by pyrolysis step (2.4) as Reaction Path 3.
- Excellent candidate when cost of HCl is low
- Solution for specific problem denoted as **Alternative 3**.

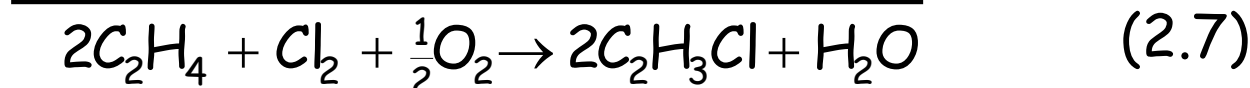
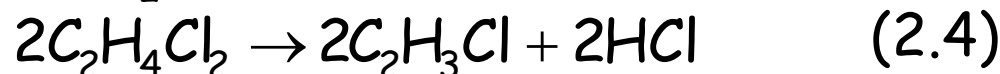
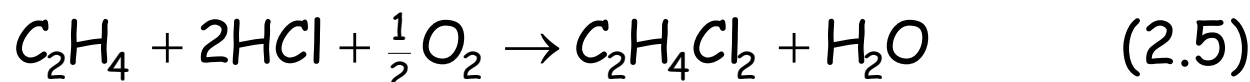
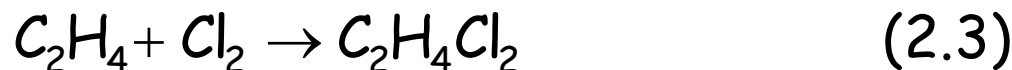
Disadvantages:

- Economics dependent on cost of HCl

Selection of pathway to VCM



5 Balanced Process for Chlorination of Ethylene:



Advantages:

- Combination of Reaction Paths 3 and 4 - addresses Alternative 2.
- All Cl_2 converted to VC
- No by-products!

Evaluation of Alternative Pathways



- Reaction Path 1 is eliminated due its low selectivity
- This leaves four alternative paths, to be compared first in terms of Gross Profit

Chemical Bulk Prices

Chemical	Cost (cents/lb)
Ethylene	18
Acetylene	50
Chlorine	11
Vinyl chloride	22
Hydrogen chloride	18
Water	0
Oxygen (air)	0

Source Ref: Seider et al., 2010

Computing Gross Profit



Reaction path ③	C_2H_4	+	Cl_2	=	C_2H_3Cl	+	HCl
lb-mole	1		1		1		1
Molecular weight	28.05		70.91		62.50		36.46
lb	28.05		70.91		62.50		36.46
lb/lb of vinyl chloride	0.449		1.134		1		0.583
cents/lb	18		11		22		18

Gross profit = $22(1) + 18(0.583) - 18(0.449) - 11(1.134) = 11.94$ cents/lb VC

Reaction Path	Overall Reaction	Gross Profit (cents/lb of VC)
②	$C_2H_2 + HCl = C_2H_3Cl$	-9.33
③	$C_2H_4 + Cl_2 = C_2H_3Cl + HCl$	11.94
④	$C_2H_4 + HCl + \frac{1}{2}O_2 = C_2H_3Cl + H_2O$	3.42
⑤	$2C_2H_4 + Cl_2 + \frac{1}{2}O_2 = 2C_2H_3Cl + H_2O$	7.68

Source Ref: Seider et al., 2010

Discussion

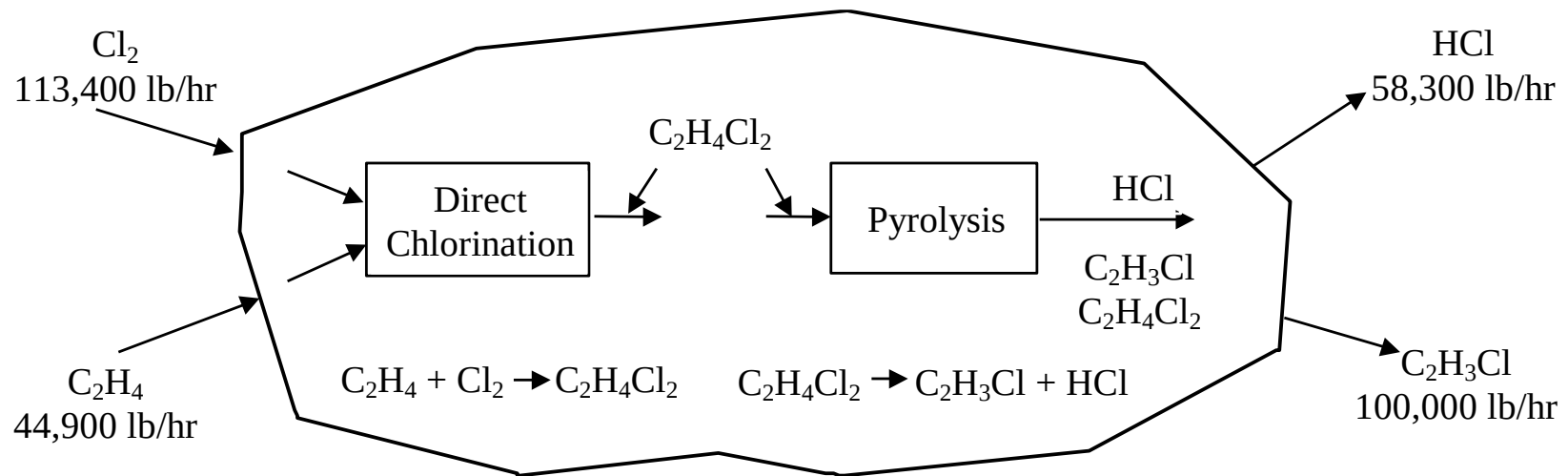


- The price of HCl strongly influences the gross profits of reaction paths 3 & 4, with the gross profit of reaction path 5 midway between the two
- Before proceeding to synthesis, the design team would be advised to examine how the gross profits vary with the price of HCl

Preliminary Flowsheet for Path - 3



- 800 MM lb/year @ 330 days/y $\Rightarrow$ 100,000 lb/hr VC
- On the basis of this principal *sink*, the HCl *sink* and reagent *sources* can be computed (each flow is 1,600 lbmol/h)



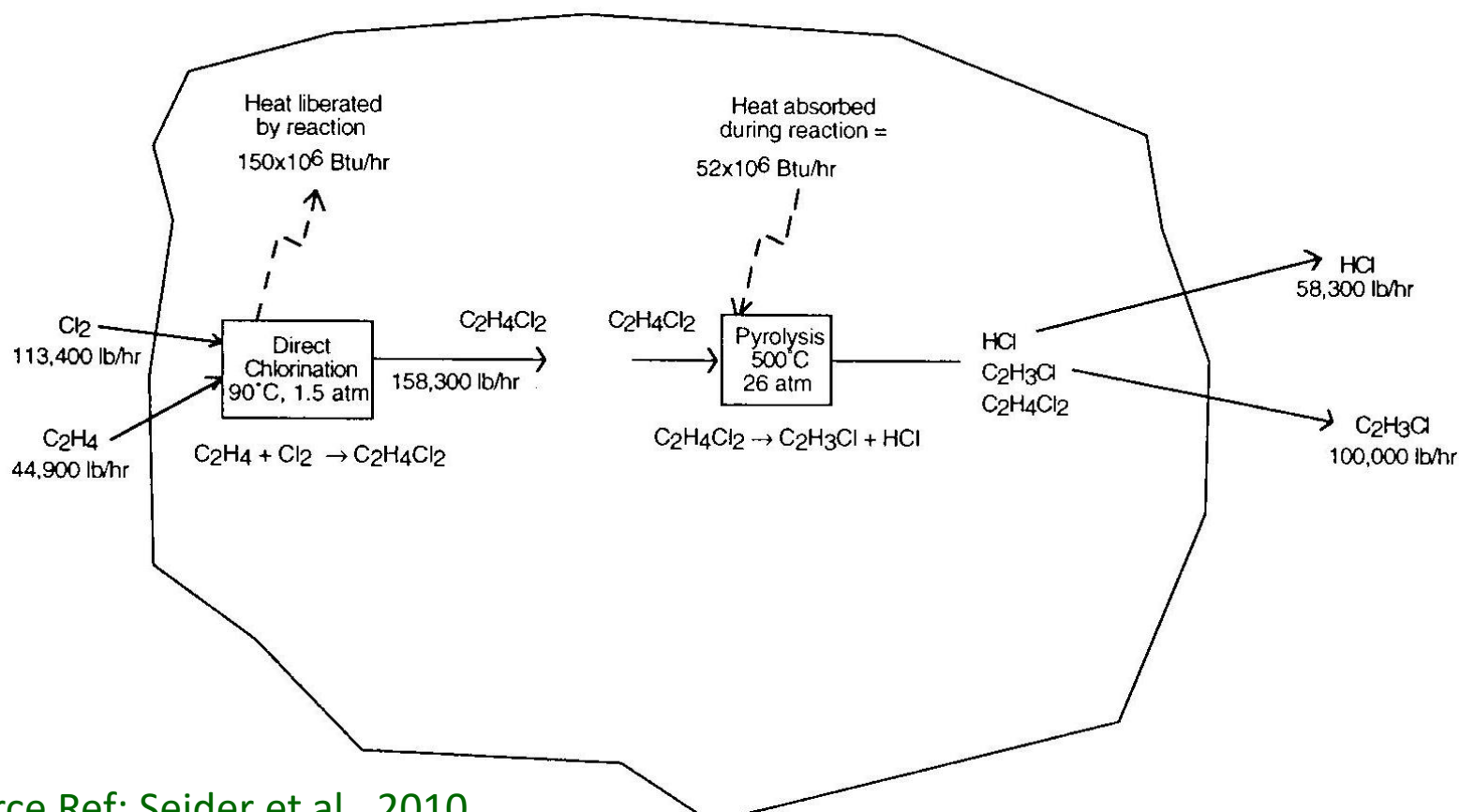
- Next step involves distributing the chemicals by matching *sources* and *sinks*.

Source Ref: Seider et al., 2010

Distribute the chemicals



- A conversion of 100% of the C_2H_4 is assumed in the chlorination reaction



Source Ref: Seider et al., 2010

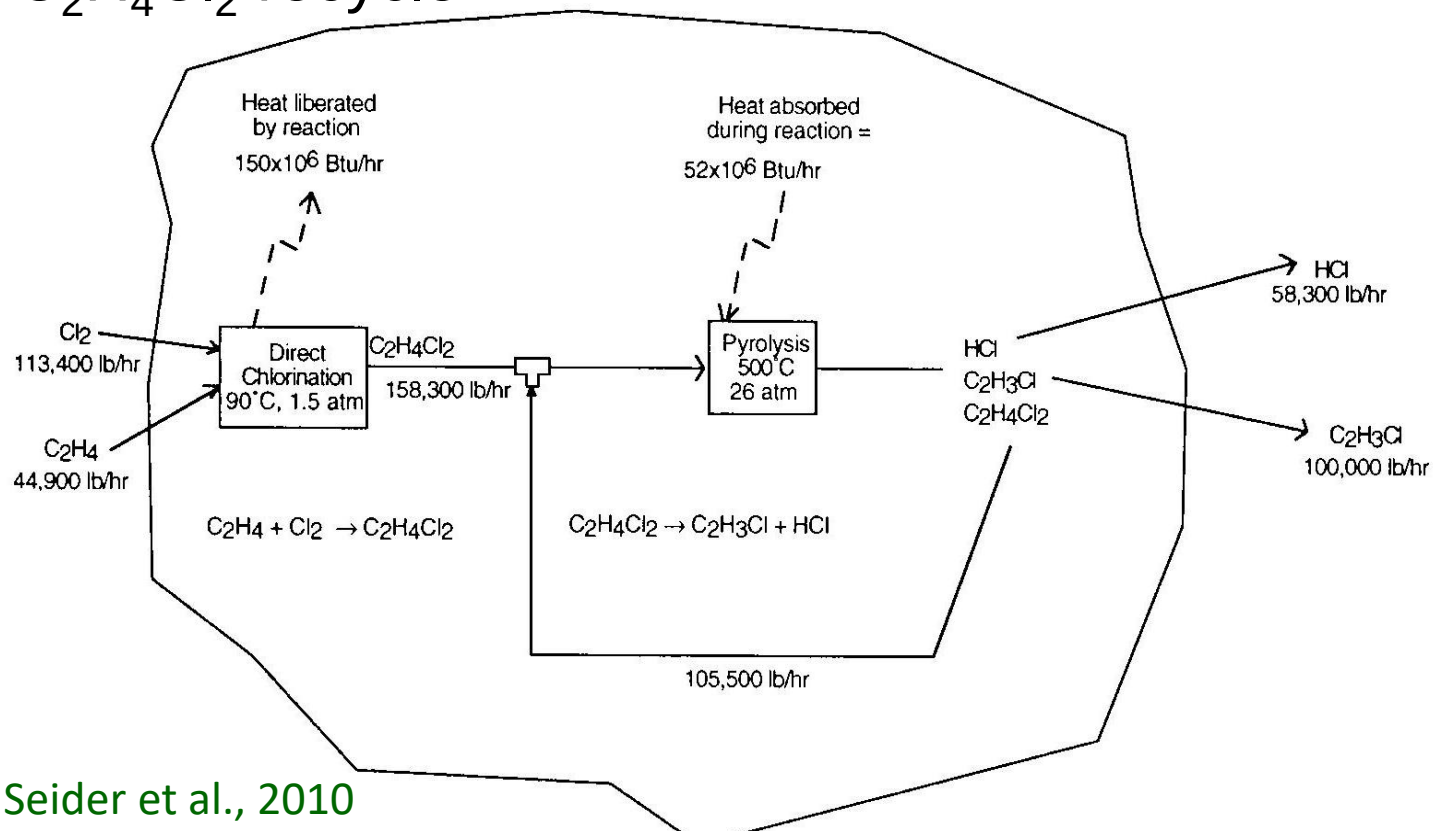
Distribute the chemicals

- Only 60% of the $\text{C}_2\text{H}_4\text{Cl}_2$ is converted to $\text{C}_2\text{H}_3\text{Cl}$ with a byproduct of HCl, according to Eqn. (2.4).
- To satisfy the overall material balance, 158,300 lb/h of $\text{C}_2\text{H}_4\text{Cl}_2$ must produce 100,000 lb/h of $\text{C}_2\text{H}_3\text{Cl}$ and 58,300 lb/h of HCl.
- But a 60% conversion only produces 60,000 lb/h of VC.
- The additional $\text{C}_2\text{H}_4\text{Cl}_2$ needed is computed by mass balance to equal:
$$[(1 - 0.6)/0.6] \times 158,300 \text{ or } 105,500 \text{ lb/h.}$$
- Its source is a recycle stream from the separation of $\text{C}_2\text{H}_3\text{Cl}$ from unreacted $\text{C}_2\text{H}_4\text{Cl}_2$, from a mixing operation, inserted to combine the two sources, to give a total 263,800 lb/h.

Distribute the chemicals



- The effluent stream from the pyrolysis operation is the source for the C_2H_3Cl product, the HCl by-product, and the $C_2H_4Cl_2$ recycle.



Source Ref: Seider et al., 2010

- Heat of reaction for the two reaction steps
 - Computed at the temperatures and pressures of the reaction operations from heats of formation and the heat capacities as a function of temperature
 - The chlorination reaction provides a large source of energy, 150 million Btu/hr, but at a lower temperature, 90°C
 - Whereas, the pyrolysis operation requires much less energy, 52 million Btu/hr, at an elevated temperature, 500°C
 - Heat source cannot be used to provide the energy for pyrolysis

- Reactor pressure levels:
 - Chlorination reaction: 1.5 atm is recommended, to eliminate the possibility of an air leak into the reactor containing ethylene.
 - Pyrolysis reaction: 26 atm is recommended by the B.F. Goodrich patent (1963) without any justification. Since the reaction is irreversible, the elevated pressure does not adversely affect the conversion. Most likely, the patent recommends this pressure to reduce the size of the pyrolysis furnace, although the tube walls must be considerably thicker and many precautions are necessary for operation at elevated pressures.
 - The pressure level is also an important consideration in selecting the separation operations, as will be discussed in the next synthesis step.

Eliminate Differences in Composition



- The product of the chlorination reaction is nearly pure $C_2H_4Cl_2$, and requires no purification.
- In contrast, the pyrolysis reactor conversion is only 60%, and one or more separation operations are required to match the required purities in the C_2H_3Cl and HCl sinks.
- One possible arrangement is given in the next slide. The data below explains the design decisions made.

Chemical	Boiling point ($^{\circ}C$)				Critical constants	
	1 atm	4.8 atm	12 atm	26 atm	$T_c, ^{\circ}C$	P_c, atm
HCl	-84.8	-51.7	-26.2	0	51.4	82.1
C_2H_3Cl	-13.8	33.1	70.5	110	159	56
$C_2H_4Cl_2$	83.7	146	193	242	250	50

Discussion



- Which separation operation? And why?
 - Distillation: because of large volatility differences
- What should be the operating conditions?
- If 1 atm, HCl boiling point -84.8°C
 - Very costly refrigeration would be necessary to condense the reflux
- At 26 atm (the pyrolysis reaction pressure), HCl boils at 0°C
 - Much less costly refrigeration could be used

Discussion



- However, recommended conditions:
 - 12 atm in which HCl boils at -26.2°C
 - And bottom products has bubble point of 93°C
 - Which is further away from the critical points of vinyl chloride-dichloroethane
 - May be selected lower pressure to avoid operation in critical region

Discussion



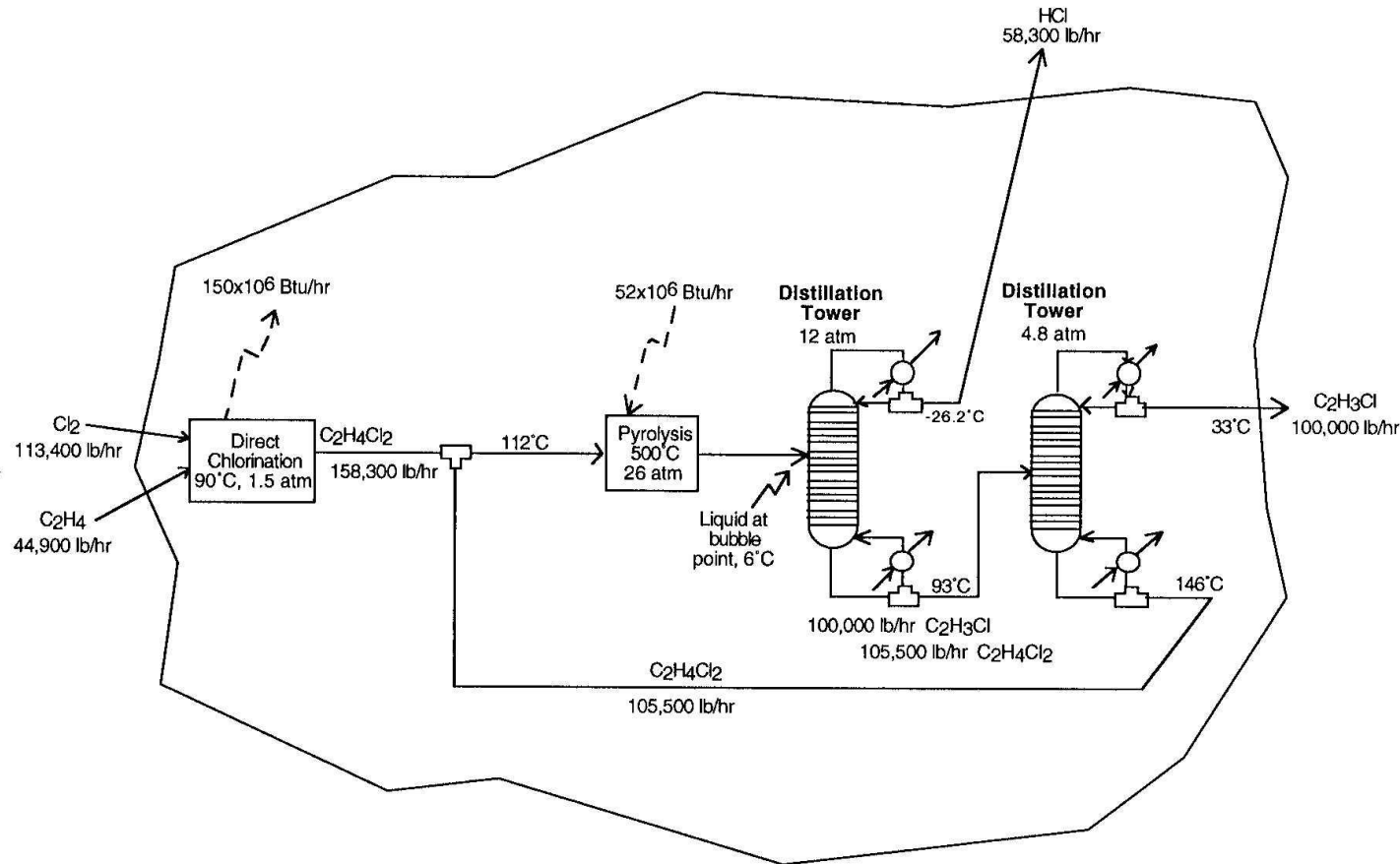
- What should be the condition of feed?
- If the feed is saturated liquid, the temperature is 6°C and 12 atm, with mild refrigerant requirement for cooling
 - A preferred feed temperature would be 35°C or higher, which could be achieved by completing the cooling and partial condensation of the pyrolysis reactor effluent with cooling water
 - However, the introduction of vapor into the column would increase the refrigeration load of the condenser at -26.2°C

Discussion



- The condition for next distillation column are specified at 4.8 atm
- The distillate boils at 33°C and can be condensed with inexpensive cooling water which is available at 25°C
- The bottom products boils at 146°C, and hence, the vapor boilup can be generated with medium-pressure steam

Eliminate Differences in Composition



- There may be other, possibly better alternative configurations

Source Ref: Seider et al., 2010

Eliminate differences in T, P and phase



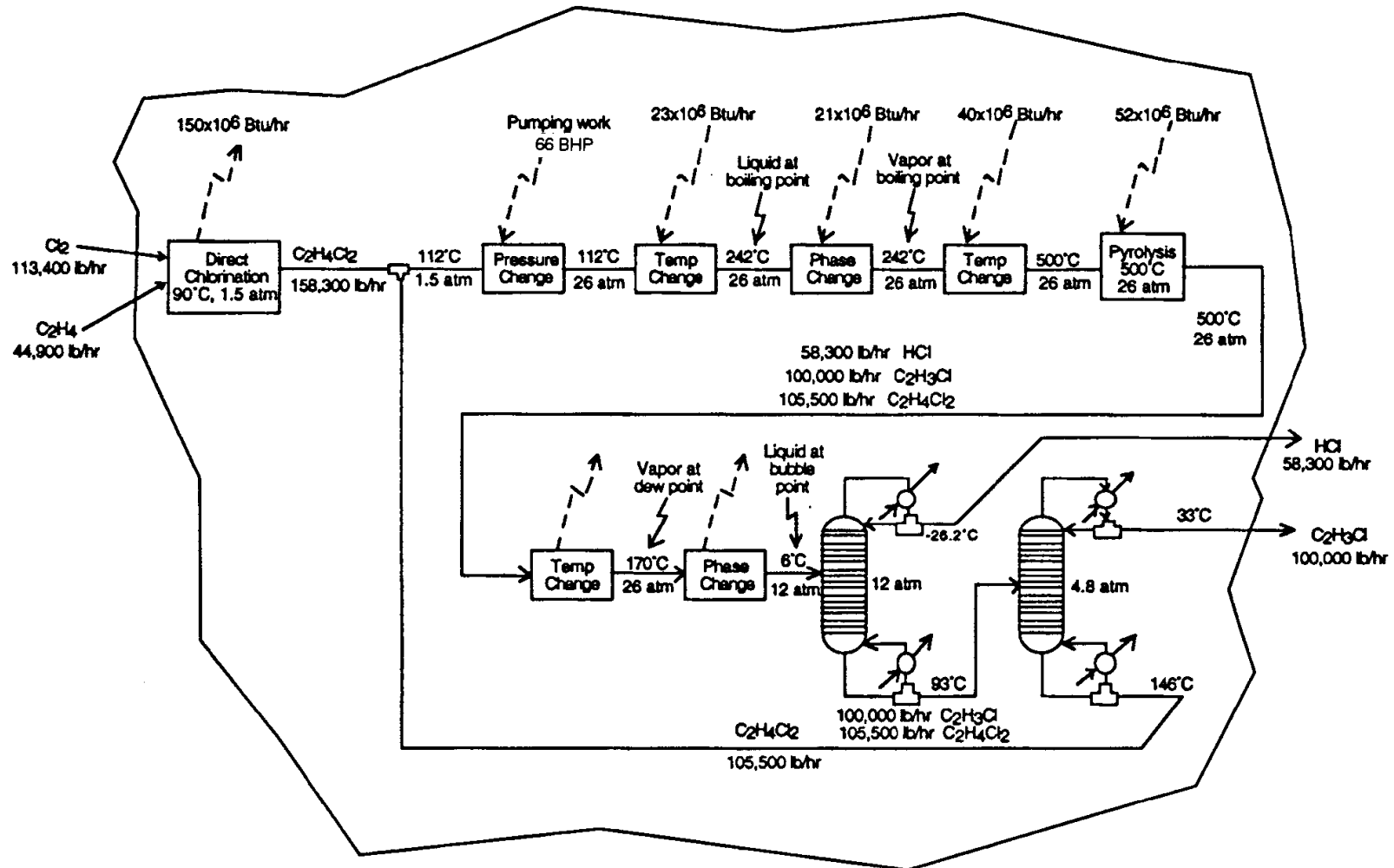
- Once the reaction and separation operations are positioned
 - The state of their feed and product streams are selected
- Liquid dichloroethane from the recycle mixer at 112°C and 1.5 atm undergoes the following operations
 - Pressure is increased to 26 atm
 - Temperature is raised to boiling point 242°C
 - Liquid is vaporized at 242°C
 - Temperature raised to pyrolysis temp, 500°C
- Alternative flow sheet would place operations 1 and 2 after operation 3.

Eliminate differences in T, P and phase



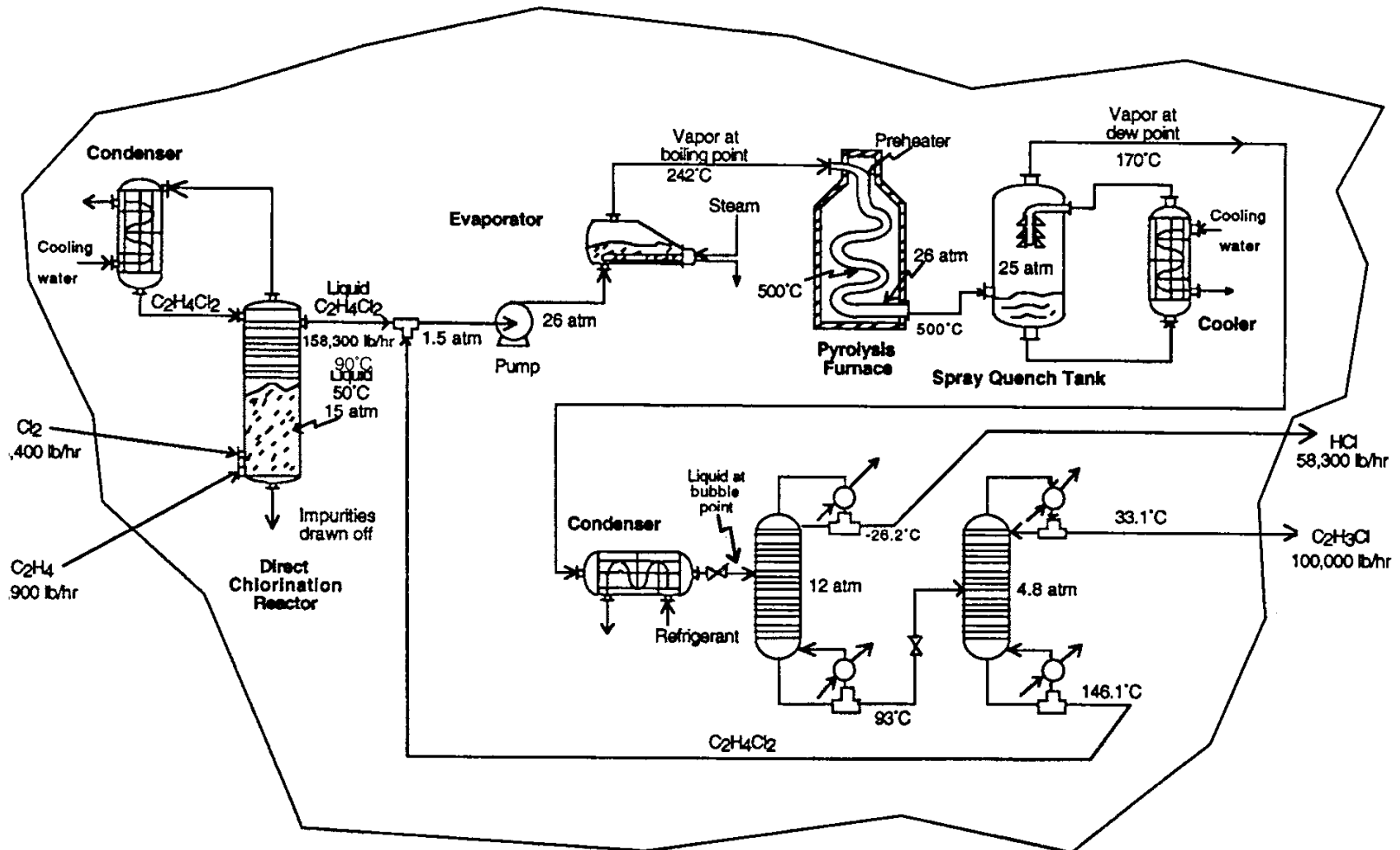
- However, this is very uneconomical
 - As the cost of compressing a vapor is so much greater than that of a liquid
 - Because the molar volume of a vapor is so much greater than that of a liquid
- Hot vapor effluent from the pyrolysis operation (at 500°C and 26 atm) is operated upon as follows
 - It's temperature is lowered to its dew point, 170°C at 26 atm
 - The vapor mixture is condensed to a liquid at its bubble point, 6°C at 12 atm, by lowering the pressure, cooling and removing the latent heat of condensation

Eliminate differences in T, P and phase



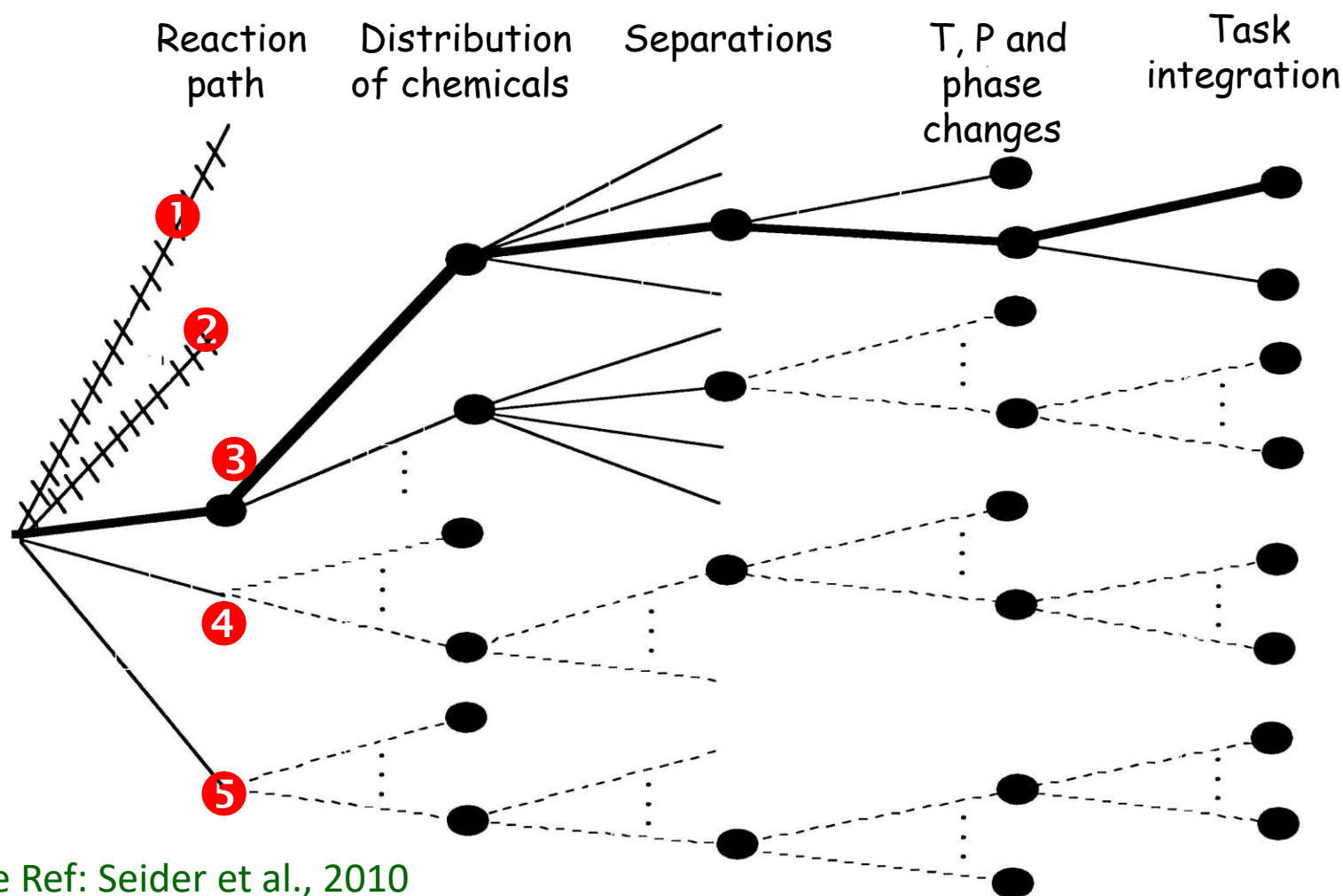
Source Ref: Seider et al., 2010

Integrate tasks (tasks → unit operations)



Source Ref: Seider et al., 2010

Assembly of synthesis tree



Source Ref: Seider et al., 2010

Flow diagrams

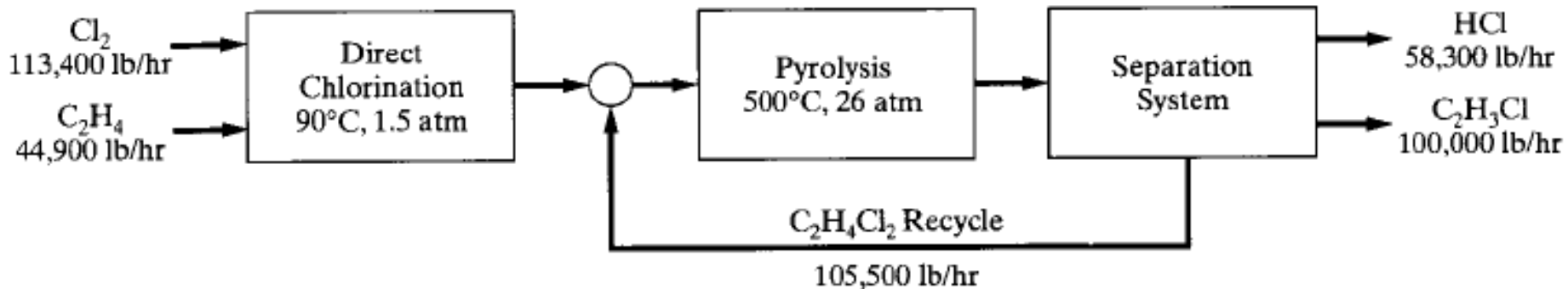


- As the engineering work on the base-case design proceeds
 - A sequence of flow diagrams is used to provide a crucial vehicle for sharing information
- The three main types are
 - Block flow diagram (BFD)
 - Process flow diagram (PFD)
 - Piping and instrumentation diagram (P&ID)

Block flow diagram (BFD)



- BFD represents the main processing sections in terms of functional blocks
 - e.g. vinyl chloride process only three main sections in the process are shown



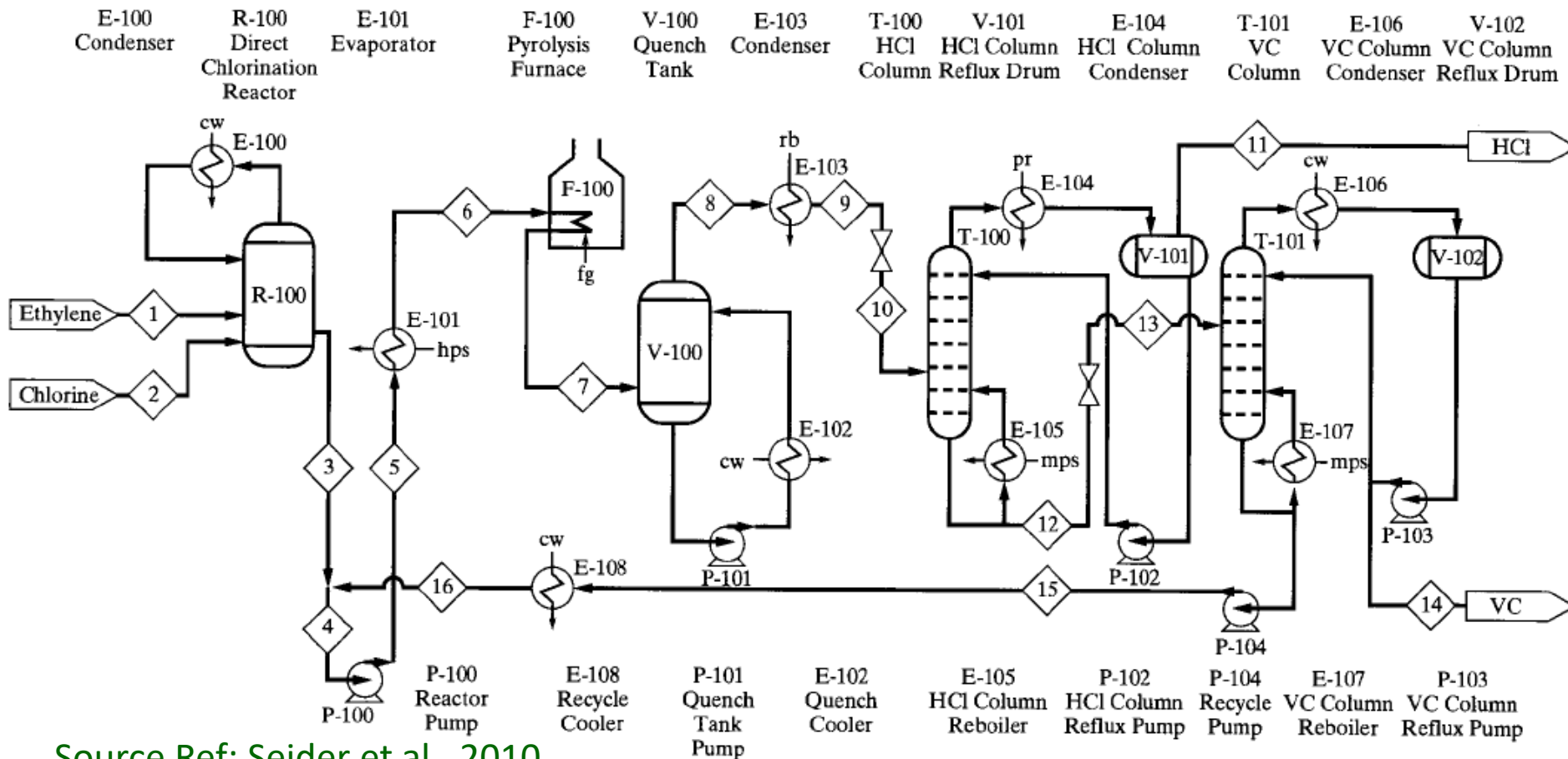
Source Ref: Seider et al., 2010

Process flow diagram (PFD)



- More detailed view of process
- Display all the major processing units in the process (including heat-exchangers, pumps, compressors, etc.)
- Provide stream information, and include the main control loops that enable the process to be regulated under normal operating conditions
- Often, preliminary PFDs are constructed using the process simulators

Process flow diagram (PFD)

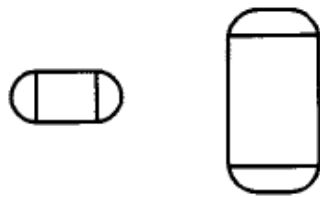


Source Ref: Seider et al., 2010

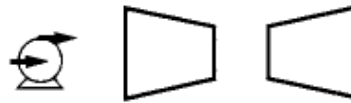
Process Units



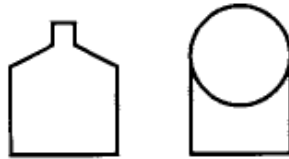
- Icons that represents the units are linked by arcs (lines) that represent the process streams
- The drawing conventions for the unit icons are taken from accepted standards



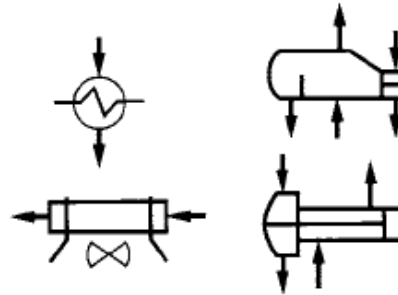
Process Vessels



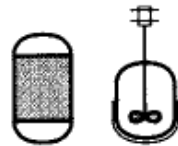
Pumps, Turbines,
Compressors



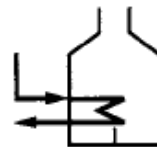
Storage Vessels



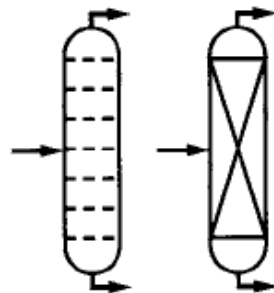
Heat Exchangers



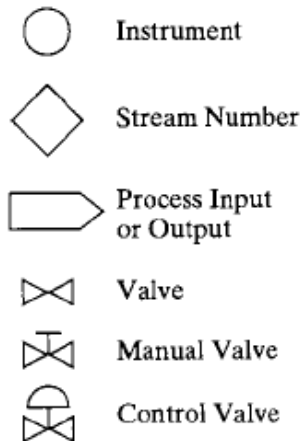
Reactors



Fired Heater



Separation Columns



Source Ref: Seider et al., 2010

Stream Information



- Directed arcs that represent the streams, with flow direction from left to right wherever possible, are numbered by reference
- Each stream is labeled on the PFD by a numbered diamond
- Furthermore, feed and product streams are identified by name

Stream Information



Table 3.6 Stream Summary Table for the Vinyl-Chloride Process in Figure 3.19

Stream Number	1	2	3	4	5	6	7	8
Temperature (°C)	25	25	90	90	91.3	242	500	170
Pressure (Atm)	1.5	1.5	1.5	1.5	26	26	26	26
Vapor fraction	1.0	1.0	0.0	0.0	0.0	1.0	1.0	1.0
Mass flow (lb/hr)	44,900	113,400	158,300	263,800	263,800	263,800	263,800	263,800
Molar flow (lbmol/hr)	1,600	1,600	1,600	2,667	2,667	2,667	4,267	4,267
Component molar flow (lbmol/hr):								
Ethylene	1,600	0	0	0	0	0	0	0
Chlorine	0	1,600	0	0	0	0	0	0
1,2-dichloroethane	0	0	1,600	2,667	2,667	2,667	1,067	1,067
Vinyl chloride	0	0	0	0	0	0	1,600	1,600
Hydrogen chloride	0	0	0	0	0	0	1,600	1,600

Source Ref: Seider et al., 2010

- Various utility streams are utilized for heating or cooling the process streams

Table 3.7 Heating and Cooling Utilities—Identifiers and Temperature Ranges

Identifier	Utility	Typical Operating Range (°F)
Hot Utilities—In increasing cost per BTU:		
<i>lps</i>	Low-pressure steam, 15 to 30 psig	250 to 275°F
<i>mps</i>	Medium-pressure steam, 100 to 150 psig	325 to 366°F
<i>hps</i>	High-pressure steam, 400 to 600 psig	448 to 488°F
<i>fg</i>	Fuel oils	
<i>fo</i>	Fuel gas	Process waste stream
<i>po</i>	Petroleum oils	Below 600°F
<i>dt</i>	Dowtherms	Below 750°F
Cold Utilities—In increasing cost per BTU:		
<i>bfw</i>	Boiler feed water	Used to raise process steam
<i>ac</i>	Air cooling	Supply at 85 to 95°F—temperature approach to process 40°F
<i>rw</i>	River water	Supply at 80 to 90°F (from cooling tower), return at 110°F
<i>cw</i>	Cooling water	Supply at 80 to 90°F (from cooling tower), return at 115 to 125°F
<i>cw</i>	Chilled water	45 to 90°F
<i>rb</i>	Refrigerated brine	0 to 50°F
<i>pr</i>	Propane refrigerant	–40 to 20°F

Source Ref: Seider et al., 2010

Equipment Summary



- This provides information for each equipment item in the PFD

Table 3.8 Equipment Summary Specifications

Equipment type	Required specification
Vessel	Height, diameter, orientation, pressure, temperature, materials of construction (MOC)
Towers	Height, diameter, orientation, pressure, temperature, number of and type of trays, height and type of packing, MOC
Pumps	Driver type, flow, suction and discharge pressures, temperature, shaft power, MOC
Compressors	Driver type, inlet flow, suction and discharge pressures, temperature, shaft power, MOC
Heat exchangers	Type, area, duty, number of shell and tube passes, for both shell and tubes: operating temperature, pressure, pressure drop, and MOC
Fired heaters	Type, tube pressure and temperature, duty, radiant and convective heat transfer area, MOC

Source Ref: Seider et al., 2010

Piping and Instrumentation Diagram (P&ID)



- This is the design document transmitted by the process design engineers to the engineers responsible for plant construction
- Also used to support the startup, operation of the process and operator training
- It contains items that do not appear in PFD, such as location and type of all measurement and control instruments, the positioning of all valves, and the size, schedule, and materials of construction of piping



Thank You